

Exam 1: Practice

The accompanying printout of the file "billboard_2021_01.csv" lists the weekly Billboard Top 10 songs from January 2021.

- `date` is the date that Billboard releases the song rankings for that week.
- `last.week` is the position on the board of the song in the previous week.
- `peak.rank` is the peak rank of the song overall, in the past and in the future.
- `weeks.on.board` is the number of weeks the song has been on Billboard Top 100 up to that date *so far*.

In R, we first run the following code and get the following output:

```
1 df <- read.csv("billboard_2021_01.csv")
2 df |> head(2)
```

A data.frame: 2 × 7

	date	rank	song	artist	last.week	peak.rank	weeks.on.board
	<chr>	<int>	<chr>	<chr>	<int>	<int>	<int>
1	2021-01-30	1	Drivers License	Olivia Rodrigo	1	1	2
2	2021-01-30	2	34+35	Ariana Grande	11	2	12

Directions: State the expected output for the following two pieces of R code.

1. `df[df$song == "Mood", "rank"]`.
2. `df |> filter(song == "All I Want For Christmas Is You") |> select(weeks.on.board)`.

Directions: Write R code that accomplishes the following tasks.

3. Return the number of different songs that appeared in this list.
4. Return which songs reached their peak position in January 2021.
5. Return the songs which ranked higher in the week of January 30th than it did in the previous week.
6. Return which songs first entered the Billboard Top 10 in January.
7. Add a column called "january_occurrences" to the data frame `df` that tabulates the number of times a given song appears in the Billboard Top 10 in January 2021.
8. Return which songs appeared exactly once (i.e. for exactly one week) on the Billboard Top 10 in January 2021.
9. Rename every occurrence of "AJR" in `df` to "Adam Jack Ryan" and save this new data frame to a data frame called `df2`. Do not overwrite `df`.
10. Add a column to `df` called "days.on.board" that reports the number of days the song had spent on the Billboard Top 100 up to that date.
11. Chart the weekly ranking of the song "Mood" with a line graph in ggplot or plotly.
12. Draw a bar chart that displays the distribution of the number of characters there are in the song titles from this list of songs (do not count the same song title twice).
13. Find the three songs in this list that jumped the most in rank from the past week, out of the songs which had been on the Billboard Top 100 the previous week.
14. Return the artists who had more than one non-collaborating song on Billboard Top 10 in January 2021.
15. Report the songs by AJR, SZA, and Ariana Grande from this list ordered by most weeks on board, with the following expected output (note the change in column names):

Song Name	Artist	Weeks on Board
<chr>	<chr>	<int>
Bang!	AJR	29
Positions	Ariana Grande	13
34+35	Ariana Grande	12
Good Days	SZA	4

date	rank	song	artist	last.week	peak.rank	weeks.on.board
1/30/2021	1	Drivers License	Olivia Rodrigo	1	1	2
1/30/2021	2	34+35	Ariana Grande	11	2	12
1/30/2021	3	Mood	24kGoldn Featuring iann dior	2	1	24
1/30/2021	4	Blinding Lights	The Weeknd	4	1	59
1/30/2021	5	Positions	Ariana Grande	3	1	13
1/30/2021	6	Levitating	Dua Lipa Featuring DaBaby	6	6	16
1/30/2021	7	Go Crazy	Chris Brown & Young Thug	5	5	37
1/30/2021	8	Holy	Justin Bieber Featuring Chance The Rapper	7	3	18
1/30/2021	9	Bang!	AJR	8	8	29
1/30/2021	10	Good Days	SZA	10	10	4
1/23/2021	1	Drivers License	Olivia Rodrigo	NA	1	1
1/23/2021	2	Mood	24kGoldn Featuring iann dior	1	1	23
1/23/2021	3	Positions	Ariana Grande	2	1	12
1/23/2021	4	Blinding Lights	The Weeknd	3	1	58
1/23/2021	5	Go Crazy	Chris Brown & Young Thug	5	5	36
1/23/2021	6	Levitating	Dua Lipa Featuring DaBaby	7	6	15
1/23/2021	7	Holy	Justin Bieber Featuring Chance The Rapper	4	3	17
1/23/2021	8	Bang!	AJR	9	8	28
1/23/2021	9	Wasted On You	Morgan Wallen	NA	9	1
1/23/2021	10	Good Days	SZA	23	10	3
1/16/2021	1	Mood	24kGoldn Featuring iann dior	1	1	22
1/16/2021	2	Positions	Ariana Grande	2	1	11
1/16/2021	3	Blinding Lights	The Weeknd	3	1	57
1/16/2021	4	Holy	Justin Bieber Featuring Chance The Rapper	4	3	16
1/16/2021	5	Go Crazy	Chris Brown & Young Thug	6	5	35
1/16/2021	6	Anyone	Justin Bieber	NA	6	1
1/16/2021	7	Levitating	Dua Lipa Featuring DaBaby	10	7	14
1/16/2021	8	Laugh Now Cry Later	Drake Featuring Lil Durk	7	2	21
1/16/2021	9	Bang!	AJR	11	9	27
1/16/2021	10	I Hope	Gabby Barrett Featuring Charlie Puth	8	3	54
1/9/2021	1	Mood	24kGoldn Featuring iann dior	7	1	21
1/9/2021	2	Positions	Ariana Grande	14	1	10
1/9/2021	3	Blinding Lights	The Weeknd	NA	1	56
1/9/2021	4	Holy	Justin Bieber Featuring Chance The Rapper	21	3	15
1/9/2021	5	Dynamite	BTS	44	1	19
1/9/2021	6	Go Crazy	Chris Brown & Young Thug	24	6	34
1/9/2021	7	Laugh Now Cry Later	Drake Featuring Lil Durk	25	2	20
1/9/2021	8	I Hope	Gabby Barrett Featuring Charlie Puth	30	3	53
1/9/2021	9	All I Want For Christmas Is You	Mariah Carey	1	1	44
1/9/2021	10	Levitating	Dua Lipa Featuring DaBaby	27	10	13
1/2/2021	1	All I Want For Christmas Is You	Mariah Carey	2	1	43
1/2/2021	2	Rockin' Around The Christmas Tree	Brenda Lee	3	2	38
1/2/2021	3	Jingle Bell Rock	Bobby Helms	4	3	35
1/2/2021	4	A Holly Jolly Christmas	Burl Ives	6	4	19
1/2/2021	5	It's The Most Wonderful Time Of The Year	Andy Williams	7	5	20
1/2/2021	6	Feliz Navidad	Jose Feliciano	10	6	14
1/2/2021	7	Mood	24kGoldn Featuring iann dior	5	1	20
1/2/2021	8	Let It Snow, Let It Snow, Let It Snow	Dean Martin	11	8	13
1/2/2021	9	Last Christmas	Wham!	14	9	19
1/2/2021	10	Run Rudolph Run	Chuck Berry	29	10	8